

ITI - Bank Queuing System

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1 Introduction

The ITI Bank Queuing System (ITI_BQS) is an embedded digital system designed to monitor and display the status of the client queue in front of bank tellers. The system detects clients entering and leaving the queue using photocells placed at the two ends of the queue.

The main purpose of ITI_BQS is to determine the current number of clients waiting in the queue, *Pcount*, and calculate the expected waiting time, *Wtime*, based on the number of available tellers. The queue status is also represented using binary empty and full flags. In addition, the system generates an alarm when a client attempts to leave an already empty queue or enter an already full queue.

To ensure that a client is counted only once even if the photocell beam remains interrupted for a period of time, edge detectors are used to convert the photocell signals into single-cycle pulses. An n-bit up-down counter is then used to maintain the number of clients in the queue. The waiting time is implemented using a ROM lookup table instead of combinational arithmetic to provide a simple and efficient hardware implementation.

The system was designed using a modular architecture in Verilog HDL. The design is divided into several components, including edge detectors, an up-down counter, a controller FSM, a ROM for the waiting-time calculation, and seven-segment display decoders. The complete system is verified through simulation using a dedicated Verilog testbench covering normal queue operation, empty and full conditions, alarm generation, and counter boundary conditions.

2 Ports

The main input and output signals of the Bank queuing system are summarized in the following table.

Signal	Direction	Width	Description
CLK	Input	1	Internal system clock.
RST	Input	1	Resets the system, clears <i>Pcount</i> , and sets the queue to empty.
Enter_Cell	Input	1	Photocell signal at the back end of the queue used to detect entering clients.
Leave_Cell	Input	1	Photocell signal at the front end of the queue used to detect leaving clients.
T1	Input	1	Indicates whether teller 1 is available.
T2	Input	1	Indicates whether teller 2 is available.
T3	Input	1	Indicates whether teller 3 is available.
Pcount	Output	3	Number of clients currently waiting in the queue, from 0 to 7.
Tcount	Output	2	Number of available tellers, from 0 to 3.
Wtime	Output	7	Expected waiting time in seconds.
Pcount_SS	Output	7	Seven-segment encoded value for displaying <i>Pcount</i> .
Tcount_SS	Output	7	Seven-segment encoded value for displaying <i>Tcount</i> .
Wtime_tens_SS	Output	7	Seven-segment encoded value for the tens digit of <i>Wtime</i> .
Wtime_unit_SS	Output	7	Seven-segment encoded value for the units digit of <i>Wtime</i> .
FULL_QUEUE	Output	1	Indicates that the queue is full.
EMPTY_QUEUE	Output	1	Indicates that the queue is empty.
FULL_ALARM	Output	1	Indicates an attempt to enter when the queue is full.
EMPTY_ALARM	Output	1	Indicates an attempt to leave when the queue is empty.

Table 1: ITI_BQS input and output ports

3 System Architecture

The ITI_BQS is designed using a modular architecture. The system is divided into several small modules, where each module performs a specific function. This improves the organization of the design and allows modules to be reused.

The main modules of the system are:

- Edge Detectors for detecting client entry and exit.
- Controller FSM for controlling the queue operation.
- Up-Down Counter for generating $Pcount$.
- ROM for storing the calculated $Wtime$ values.
- Seven-Segment Decoders for displaying the system information.

3.1 Block Diagram

The overall structure of the ITI_BQS is shown in Figure 1.

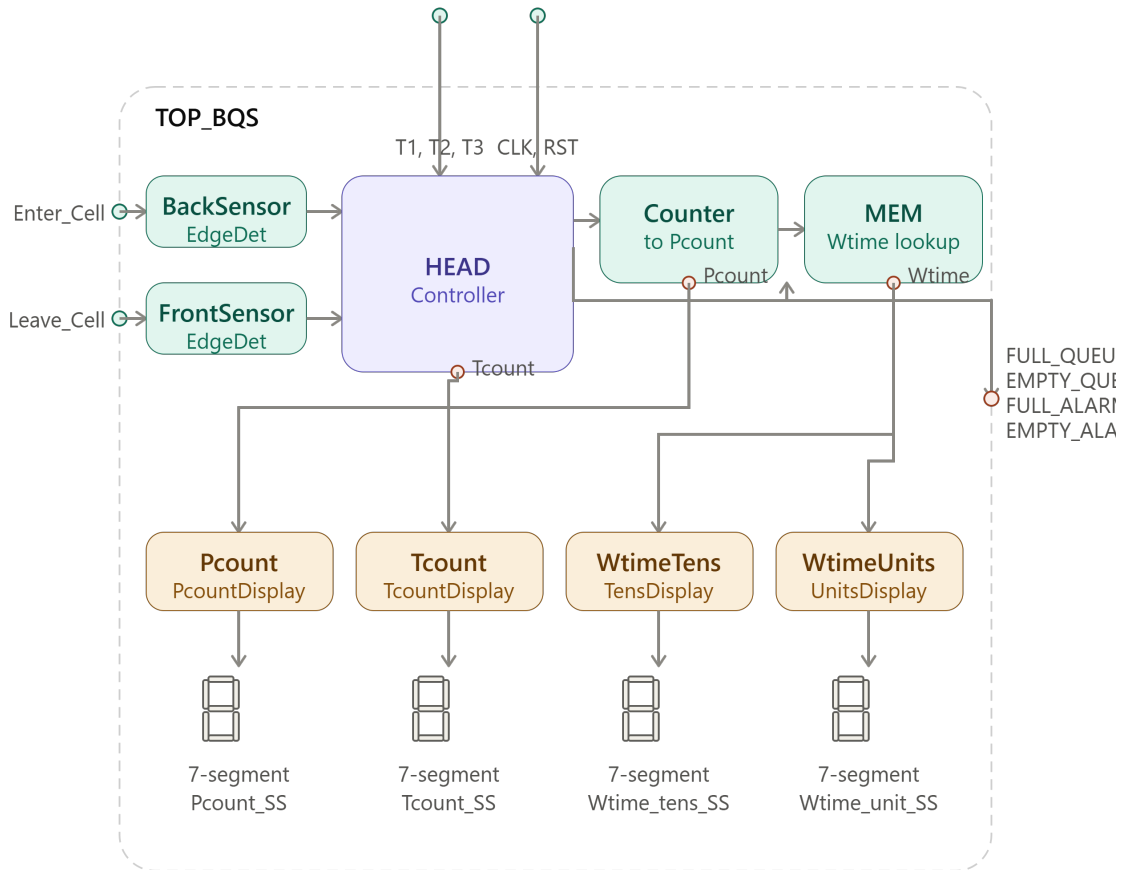


Figure 1: ITI_BQS block diagram

4 FSM Design

The queue operation is controlled by a finite state machine (FSM). The FSM has three states: **EMPTY**, **COUNT**, and **FULL**. The current state depends on the number of clients in the queue and the detected entry or exit events.

- **EMPTY**: The queue contains no clients ($Pcount = 0$).
- **COUNT**: The queue contains between 1 and 6 clients.
- **FULL**: The queue has reached its maximum capacity ($Pcount = 7$).

The FSM generates the control signals for the up-down counter and the queue status flags. It also generates the appropriate alarm when an entry is attempted while the queue is full or an exit is attempted while the queue is empty.

4.1 FSM State Diagram

The state transitions of the controller are shown in Figure 2.

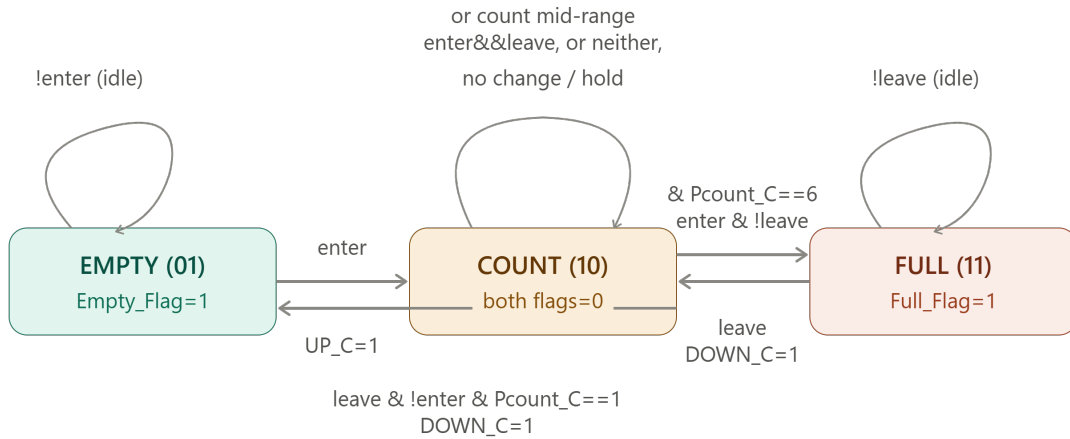


Figure 2: ITI_BQS controller FSM

5 System Modules

The ITI_BQS system is implemented using several Verilog modules. Each module is responsible for a specific part of the system operation, which makes the design modular and easier to verify.

5.1 Edge Detector

The Edge Detector is used to detect a new client entering or leaving the queue. Since the photocell signal may remain active for more than one clock cycle, the Edge Detector generates a single-cycle pulse when a valid transition is detected.

Two Edge Detector modules are used in the system: one for the entering photocell and another for the leaving photocell.

5.2 Up-Down Counter

The Up-Down Counter maintains the current number of clients in the queue, represented by $Pcount$. The counter increments when a client enters the queue and decrements when a client leaves.

The counter operates within the range from 0 to 7. The FSM controls the counting operation to prevent the counter from exceeding the maximum queue capacity or becoming negative.

5.3 Waiting-Time ROM

The expected waiting time, $Wtime$, is stored in a ROM lookup table. The ROM uses the combination of $Pcount$ and $Tcount$ as its address.

The waiting time is calculated according to:

$$Wtime = \frac{3(Pcount + Tcount - 1)}{Tcount}$$

The calculated values are stored in the ROM so that the system can obtain the required waiting time directly without performing the division operation during normal operation.

When $Tcount = 0$, the waiting time is considered undefined.

5.4 Seven-Segment Display

Seven-segment decoder modules are used to display the main system information. The displayed values include:

- Number of clients in the queue, $Pcount$.
- Number of available tellers, $Tcount$.
- Tens digit of $Wtime$.
- Units digit of $Wtime$.

5.5 Queue Status and Alarms

The system provides status signals to indicate whether the queue is empty or full.

- **EMPTY_QUEUE**: Asserted when $Pcount = 0$.
- **FULL_QUEUE**: Asserted when $Pcount = 7$.
- **EMPTY_ALARM**: Generated when a leaving event occurs while the queue is empty.
- **FULL_ALARM**: Generated when an entering event occurs while the queue is full.

6 System Operation

The system operation starts after reset, where the queue is initialized to the empty state and $Pcount$ is set to zero.

When a client enters the queue, the entering photocell detects the event and the Edge Detector generates a single-cycle pulse. The FSM then enables the counter to increment $Pcount$.

Similarly, when a client leaves the queue, the leaving photocell generates a pulse and the FSM enables the counter to decrement $Pcount$.

The FSM also handles the boundary conditions. If the queue is empty and a leaving event occurs, the counter is not decremented and the **EMPTY_ALARM** signal is generated. If the queue is full and another entering event occurs, the counter remains at 7 and the **FULL_ALARM** signal is generated.

The number of available tellers is continuously determined from the teller inputs. The corresponding waiting time is then obtained from the ROM and displayed on the seven-segment displays.

7 Verification and Simulation

The ITI_BQS design was verified using a Verilog testbench. Different operating conditions were applied to verify the functionality of the individual modules and the complete system.

The testbench covers the following cases:

- Reset operation.
- Client entry and exit.
- Counting from an empty queue to a full queue.
- Counting down from a full queue.
- Empty queue condition.
- Full queue condition.
- Empty queue alarm.
- Full queue alarm.
- Different numbers of available tellers.
- Waiting-time ROM values.

The simulation waveforms were checked to ensure that the FSM transitions, *Pcount*, *Tcount*, *Wtime*, queue status signals, and alarm signals operate according to the system specification.

7.1 Simulation Results

The simulation results demonstrate the correct operation of the ITI_BQS system under the tested conditions. The queue counter changes according to the detected entering and leaving events, while the FSM changes between the **EMPTY**, **COUNT**, and **FULL** states according to the queue count.

The waiting-time output also changes according to the current values of *Pcount* and *Tcount*, while the appropriate queue status and alarm signals are generated at the boundary conditions.

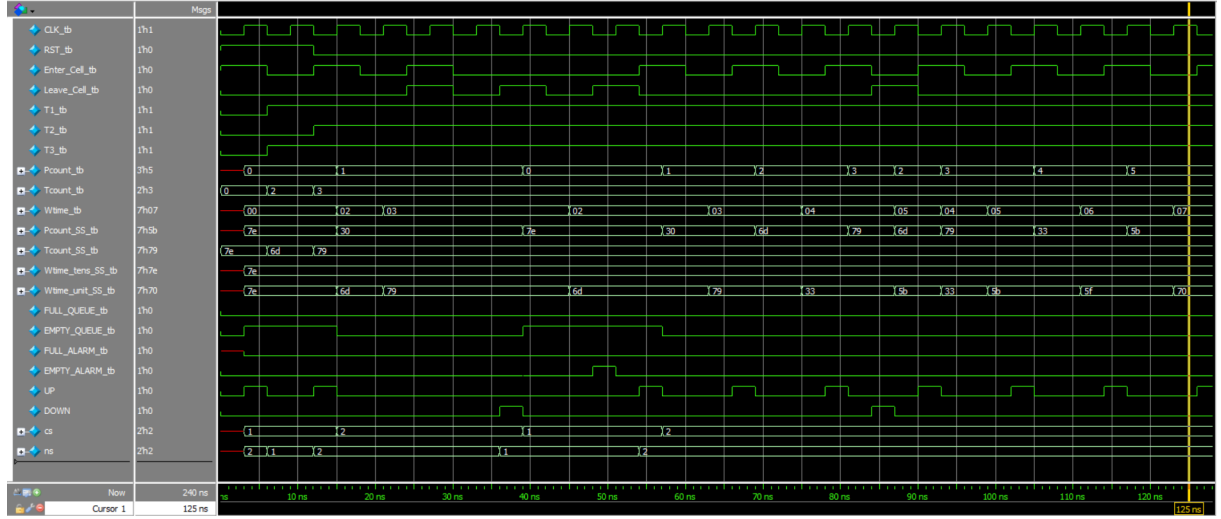


Figure 3: BQS simulation waveform

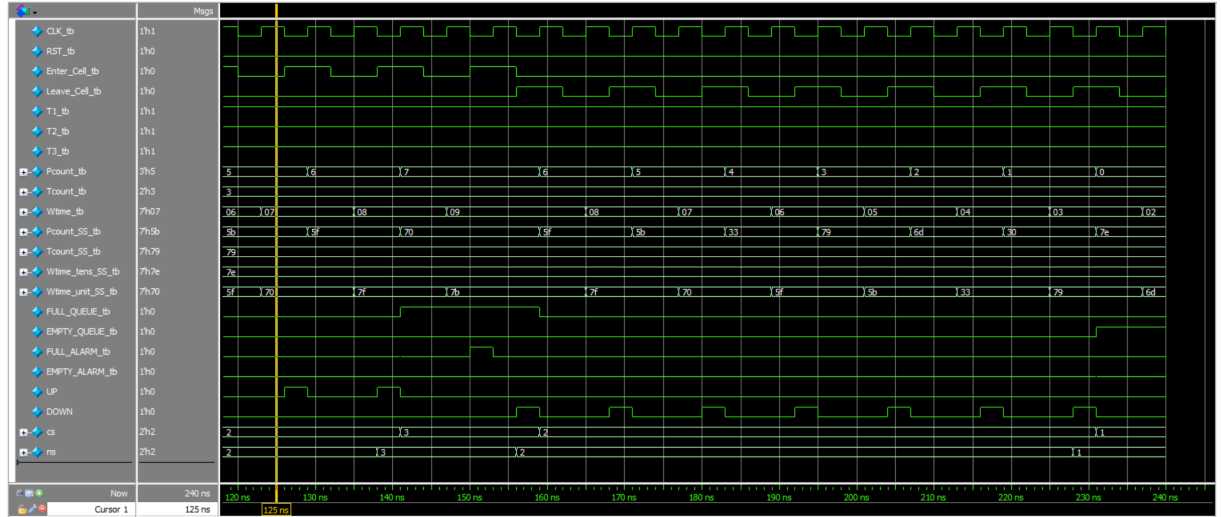


Figure 4: BQS simulation waveform

8 Conclusion

The ITI Bank Queuing System was successfully designed using Verilog HDL with a modular architecture. The system detects client entry and exit, maintains the number of clients in the queue, determines the available tellers, and provides the expected waiting time.

The FSM manages the queue states and boundary conditions, while the Edge Detectors ensure that each client event is counted only once. The waiting-time ROM provides the required waiting-time values based on the queue size and available tellers.

The complete design was verified through simulation using different queue conditions and input combinations to ensure correct system operation.